Mert Şeker

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Summary

Unity Gameplay Programmer with 5+ years of engineering experience in real-time systems, simulation, and control engineering. Developed Unity projects featuring procedural generation, multiplayer gameplay, persistence systems, and Android/WebGL deployment. Strong foundation in gameplay architecture, debugging, and performance-sensitive systems.

Experience

Control Systems Engineer Jan 2019 – Feb 2024
ASELSAN A.Ş. *Ankara, Turkey*
- Developed real-time drive and control systems for robotic weapon platforms using MATLAB and Simulink.
- Created and simulated motion-planning algorithms using Python, ROS 2, and Gazebo.
- Worked with physics-based simulations, time-step loops, state-based logic, and performance-sensitive systems.

Electrical and Electronics Engineer Jul 2018 – Jan 2019
Otonom Teknoloji Robotics, Electronics & Software *Ankara, Turkey*
- Participated in the design and analysis of embedded systems and PCB layouts for robotic systems.

Projects

Circuit Repair – Procedural Puzzle Game | Unity, C#, PCG, Android 2026
- Developed a mobile-first circuit rotation puzzle featuring deterministic procedural generation using randomized backtracking, Hamiltonian paths, and constraint pruning.
- Implemented seed-based puzzle generation, difficulty filtering, reproducible scrambling, and automated validation across 1,000 generated puzzles.
- Designed a modular gameplay architecture separating generation, evaluation, input, content, and presentation systems; tested the game on Android.

Core Charge – Idle / Incremental Game | Unity, C#, WebGL 2026
- Built and published a short sci-fi idle/incremental game prototype on itch.io, with generator progression, milestone multipliers, upgrade layers, and prestige.
- Implemented a tick-based economy simulation, buy modes, unlock requirements, Overflow upgrades, and Core Reactor prestige progression.
- Built save/load and offline progress systems using timestamp-based calculations and browser-compatible persistence for WebGL.
- Implemented player-facing UI flows including tutorial, settings, audio controls, feedback popups.

Turn-Based Stock Market Board Game | Unity, C#, Mirror 2025
- Designed and implemented a 2.5D turn-based strategy board game with stock-market simulation mechanics.
- Implemented server-authoritative multiplayer features using Mirror, including turn flow, player interactions, and networked game state handling.
- Built core gameplay systems including stock price simulation, drafting logic, turn management, bidding and player decision flow.
- Used ScriptableObjects and event-based patterns to structure gameplay data, UI updates, and system communication.

Education

Barcelona Code School Jan 2025 – Mar 2025
Bootcamp: Game Development with Unity and C# *Barcelona, Spain*

Istanbul Technical University Sep 2020 – Jun 2023
M.Sc. in Mechanical Engineering, System Dynamics and Control *Istanbul, Turkey*
- Thesis: "Optimizing Dynamic Window Approach for Autonomous Robots Using Deep Neural Networks."

Middle East Technical University Sep 2011 – Jun 2017
B.Sc. in Electrical and Electronics Engineering *Ankara, Turkey*

Skills

Game Development: Unity, C#, Procedural Generation, Multiplayer (Mirror), ScriptableObjects, Android, WebGL
Programming & Tools: Python, Git, MATLAB, Simulink
Engineering: Control Systems, Motion Planning, ROS 2, Gazebo
Languages: Turkish (Native), English (Professional)